

Notes: Chevalier (AER, 1995)

Capital Structure and Product-Market Competition: Empirical Evidence from the Supermarket Industry

Contents

1	Big picture	3
2	Data and measurement (key objects)	3
2.1	Industry setting and LBO activity	3
2.2	Event-study sample	4
2.3	Local market panel for entry, exit, and expansion	4
2.4	Constructed variables	5
2.5	Treatment of post-LBO asset sales and mergers	5
2.6	Unobservables and timing	5
3	Models and empirical specifications (all equations + interpretation)	6
3.1	Conceptual prediction from theory	6
3.2	Event-study market model	6
3.3	Separating direct competitors from noncompetitors	7
3.4	Full-market store-growth regression	7
3.5	Ordered probit for incumbent expansion	7
3.6	OLS robustness for incumbent store changes	8
3.7	Probit for de novo entry	8
3.8	Alternative-hypothesis tests	8
4	Identification and instruments (what makes the estimates credible)	8
4.1	What fails in a naive interpretation	8
4.2	Why the event study is informative	9
4.3	Why the local-market design helps	9
4.4	Why early versus late LBOs help identification	9

4.5	Why the weak-firm alternative is not persuasive	9
4.6	Relation to the literature	10
5	Tables	10
5.1	Tables 1 and 2: event setup and rival sample	10
5.2	Table 3: rival stock-price responses to LBO announcements	10
5.3	Table 4: variable definitions and summary statistics	11
5.4	Table 5: full-market OLS regressions	11
5.5	Table 6: incumbent expansion decisions	12
5.6	Table 7: early versus late LBOs in incumbent expansion	12
5.7	Table 8: OLS robustness for incumbent store changes	13
5.8	Table 9: de novo entry into local markets	14
5.9	Table 10: are future LBO firms simply weak firms?	14
6	Short summary	15
7	Conclusion	15
7.1	Core conclusion (what the paper establishes)	15
7.2	Economic intuition	15
7.3	Takeaway	15

1 Big picture

- **Question.** Does a firm’s capital structure affect how aggressively it competes in the product market?
- **Setting.** The paper studies the U.S. supermarket industry during the late-1980s wave of leveraged buyouts (LBOs). Supermarket competition is local, so the author can compare cities with different exposure to LBO firms.
- **Core challenge.** If leverage changes product-market competition, rivals should react. But LBOs may also be correlated with other forces: takeover speculation, weak-firm selection, or simple asset reshuffling. The empirical design therefore has to separate *competition effects* from *selection effects*.
- **Main test 1.** An event study asks whether the stock price of a supermarket chain rises or falls when a rival announces an LBO.
- **Main test 2.** A local-market analysis asks whether markets with more LBO incumbents experience more or less *entry*, *expansion*, and *total store growth*.
- **Competing theories.** One class of models predicts that more leverage **softens** competition; another predicts that it **toughens** competition.
- **Main finding.** The evidence supports the “softer competition” view. Rival supermarket chains earn positive abnormal returns around LBO announcements, and markets with more LBO firms attract more rival entry and expansion.
- **Interpretation.** Higher leverage appears to reduce the aggressiveness of product-market competition in this industry.
- **Why the paper matters.** This is one of the early empirical papers showing that financing choices can have **real product-market effects**, not just balance-sheet effects.

2 Data and measurement (key objects)

2.1 Industry setting and LBO activity

- **Industry.** A supermarket is defined as a retail food store with annual sales above \$2 million and more than 9,000 square feet of selling space.
- **National versus local competition.** National concentration is modest, but competition is local. The four largest chains accounted for only 16% of U.S. grocery sales in 1982, yet the average metropolitan statistical area (MSA) had a four-firm concentration ratio of 58%.
- **Why local markets matter.** Supermarket rivalry is naturally organized at the MSA/division level: chains have local divisional managers, local warehouses, and local advertising circulars.
- **Wave of leverage shocks.** The key transactions occurred mainly between 1985 and 1988. The largest were:
 - Safeway LBO: \$5.3 billion,
 - Kroger leveraged recapitalization: \$4.1 billion,

- Supermarkets General LBO: \$1.8 billion,
- Stop & Shop LBO: \$1.2 billion.
- **Scale.** These four firms alone owned nearly 4,000 U.S. supermarkets at the time of their transactions.
- **Breadth.** Overall, 19 of the 50 largest supermarket chains undertook LBOs, representing about \$72 billion of the industry’s \$297 billion in supermarket sales in 1991.
- **Geography.** LBO exposure is spread across regions, though the West is especially exposed because of Safeway.

2.2 Event-study sample

- **Events studied.** The paper analyzes four major leverage events:
 - Safeway,
 - Supermarkets General,
 - Stop & Shop,
 - Kroger.
- **Event window.** For each event, the window begins 30 trading days before the first public announcement suggestive of a takeover/LBO and ends on the day the final LBO or leveraged recapitalization is announced.
- **Return data.** Daily stock returns come from CRSP and cover January 1985 through October 10, 1988.
- **Rival chains.** The event-study sample contains 13 rival supermarket chains that derive at least 80% of revenue from supermarket sales and trade continuously over the sample window.
- **Direct competitors.** Two chains are considered direct competitors if they both operate stores in at least one common MSA.
- **Data source for local competition.** Supermarket News’ *Distribution Study of Grocery Store Sales* is used to map store names to local markets, with firm identities filled in using 10-Ks and the *Retail Tenants Directory*.

2.3 Local market panel for entry, exit, and expansion

- **Main panel.** The local-market analysis uses data on supermarket chains in 85 MSAs in 1985 and 1991.
- **Market scope data.** The source is *Progressive Grocer’s Market Scope* volumes for 1986 and 1992, which provide counts of stores by chain in the largest U.S. MSAs.
- **Sample restriction.** The 85 MSAs are those that were among the largest 100 in both years and whose Census MSA boundaries did not change between 1985 and 1991.
- **Entry extension.** For de novo entry, the sample is extended to 1993 using Nexis searches because entry is slower than incumbent expansion.

- **Store universe.** The data include chain stores and the total number of independent stores in each MSA.

2.4 Constructed variables

- **LBO share of a market.** The share of stores in an MSA in 1985 owned by firms that eventually undertook an LBO by 1990.
- **Early versus late LBO share.** The LBO share is split into firms that leveraged before 1988 and firms that leveraged during or after 1988.
- **Total market growth.** The percentage change in the number of stores in the MSA between 1985 and 1991.
- **Incumbent expansion outcome.** For a firm–market pair, the dependent variable is

$$y_{im} \in \{-1, 0, 1\},$$

where $y_{im} = 1$ if the incumbent adds stores on net, 0 if it leaves store count unchanged, and -1 if it subtracts stores on net.

- **De novo entry indicator.** At the MSA level, the dependent variable equals 1 if a large chain that was not present in 1985 enters by 1993.
- **Market controls.** The regressions use changes in households, households per square mile, median income, median income squared, the share of households with income below \$10,000, a 1985 over- or under-storing measure, and a Herfindahl index of local concentration.
- **Firm controls.** The incumbent regressions also include the firm’s own market share in the MSA and its total number of stores in the full sample of MSAs.

2.5 Treatment of post-LBO asset sales and mergers

- **Key adjustment.** Stores sold in post-LBO asset sales are treated as if they were always owned by their eventual purchasers.
- **Reason.** Otherwise, asset divestitures would mechanically look like exit by the LBO firm and entry or expansion by the buyer, contaminating the interpretation.
- **Example.** When Safeway sold its Southern California division to Vons, those stores are reassigned to Vons in the baseline accounting.
- **Similar rule for mergers.** Non-LBO mergers are handled similarly, so observed changes reflect genuine expansion and contraction rather than ownership relabeling.

2.6 Unobservables and timing

- **Leverage itself is not observed for all firms.** The empirical design uses the LBO indicator as a proxy for “high debt” because detailed leverage data are unavailable for privately held chains.
- **Competition is local and multidimensional.** The paper refers to the “toughness” of product-market competition because rivalry may operate through both prices and store quality.

- **Timing matters.** Expansion by incumbents should adjust faster than de novo entry. This is why the paper separately studies incumbent expansion and new entry, and why it splits LBOs into early and late events.

3 Models and empirical specifications (all equations + interpretation)

3.1 Conceptual prediction from theory

The paper is not a structural model of supermarket competition. Instead, it uses reduced-form implications of two theoretical classes of models:

- **If leverage softens competition:**
 - rivals’ stock prices should rise at LBO announcements,
 - rival incumbents should expand,
 - nonincumbents should be more likely to enter,
 - more stores should “fit” in the local market.
- **If leverage toughens competition:**
 - rivals’ stock prices should fall,
 - rivals should contract or exit,
 - fewer stores should fit in the market.

3.2 Event-study market model

For rival firm i on date t , the estimated return equation is

$$R_{it} = \alpha_i + \beta_i R_{mt} + \sum_j \delta_{ij} D_{jt} + \varepsilon_{it},$$

where:

- R_{it} is firm i ’s stock return,
- R_{mt} is the value-weighted NYSE/AMEX market return,
- D_{jt} is an indicator for event window j ,
- δ_{ij} is the abnormal return associated with event j .
- **Interpretation.** A positive δ_{ij} for rival firms means that the LBO is expected to improve rivals’ future profits.
- **Estimation method.** The system is estimated by seemingly unrelated regressions (SUR) because error terms across supermarket chains are likely contemporaneously correlated.

3.3 Separating direct competitors from noncompetitors

The event study compares two groups:

- **Direct competitors:** firms with overlapping local markets,
- **Noncompetitors:** firms with no overlapping local markets.
- **Why this matters.** If the announcement merely raises speculation that more supermarket LBOs may occur, *all* firms in the industry could rise. But if the effect runs through product-market rivalry, only direct competitors should respond strongly.

3.4 Full-market store-growth regression

At the MSA level, the baseline regression takes the form

$$\Delta\text{Stores}_m = \alpha + \beta \text{LBOShare}_m + X'_m \gamma + \varepsilon_m,$$

where ΔStores_m is the percentage change in total stores in MSA m between 1985 and 1991, and X_m includes changes in demographic conditions and 1985 market structure.

- **Interpretation.** If $\beta > 0$, markets with more LBO exposure end up with more stores, consistent with softer competition.
- **Early/late variant.** The paper also estimates

$$\Delta\text{Stores}_m = \alpha + \beta_E \text{EarlyLBOShare}_m + \beta_L \text{LateLBOShare}_m + X'_m \gamma + \varepsilon_m.$$

This tests whether earlier LBOs have had more time to affect market structure.

3.5 Ordered probit for incumbent expansion

For incumbent firm i in market m , define a latent net-expansion index

$$y_{im}^* = Z'_{im} \theta + u_{im},$$

with observed outcome

$$y_{im} = \begin{cases} -1 & \text{if the firm subtracts stores on net,} \\ 0 & \text{if the firm leaves store count unchanged,} \\ 1 & \text{if the firm adds stores on net.} \end{cases}$$

The regressors include market growth controls, the firm's own market share, chain size, concentration, and the share of rivals that are LBO firms.

- **Key regressor.** For incumbent regressions, the LBO share is defined among the firm's *rivals*, excluding its own stores.
- **Interpretation.** A positive coefficient means the firm is more likely to add stores when its rivals are more leveraged.
- **Why ordered probit?** The dependent variable is discrete and highly concentrated at small integer changes, so treating the decision as an ordered response is more natural than using only continuous store changes.

3.6 OLS robustness for incumbent store changes

As a robustness check, the paper also estimates an OLS regression using the *actual number of stores added or subtracted* as the dependent variable, with White (1980) heteroskedasticity-robust standard errors.

- **Interpretation.** This checks whether the same pattern holds when focusing on the magnitude of expansion, not just its sign.

3.7 Probit for de novo entry

At the MSA level, the entry equation is

$$\Pr(\text{Entry}_m = 1) = \Phi(\alpha + \beta_E \text{EarlyLBOShare}_m + \beta_L \text{LateLBOShare}_m + W'_m \gamma),$$

where entry means a previously absent large chain opens at least one store by 1993.

- **Interpretation.** A positive coefficient on early LBO share means that leveraged incumbents make the market more attractive to outside chains.
- **Why only market-level regressors?** Entry is measured at the city level, so firm-specific incumbent controls cannot be included.

3.8 Alternative-hypothesis tests

The paper explicitly tests an alternative explanation: perhaps LBO firms were simply weak firms, so rivals expanded because those firms were poor competitors rather than because leverage changed behavior.

The paper addresses this in four ways:

- **Asset-sale reassignment.** Post-LBO asset sales are reassigned to purchasers, minimizing the role of visibly weak divisions.
- **Accounting comparisons.** Pre-LBO profitability and valuation ratios of future LBO and non-LBO firms are compared.
- **Event-study logic.** If LBO firms were already known to be weak and behavior did not change, rivals should not earn positive abnormal returns at announcement.
- **Early versus late timing.** If the relevant issue were chronic weakness, both expansion and entry should respond similarly even to later LBO firms. The actual timing pattern is more consistent with a competition change that unfolds after leverage rises.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in a naive interpretation

- **Naive reading 1: industrywide takeover speculation.** Rival stock prices could rise around an LBO announcement simply because investors expect more LBOs throughout the industry.

- **Naive reading 2: weak-firm selection.** Markets with more LBO firms may attract entry because those firms were already poor competitors, not because leverage changed product-market conduct.
- **Naive reading 3: mechanical asset transfers.** Post-LBO divestitures could create spurious entry and expansion if sold divisions are counted as newly entered or expanded operations.

4.2 Why the event study is informative

- The event study compares **direct competitors** with **noncompetitors**. If product-market competition is the channel, direct competitors should move more than noncompetitors.
- That is exactly what the paper finds: positive abnormal returns for competing chains, little or no response for noncompeting chains.
- This sharply weakens the “industrywide LBO speculation” story.

4.3 Why the local-market design helps

- Supermarket rivalry is local. By working at the MSA level, the paper ties the LBO shock to the local competitive environment rather than to the national firm only.
- The treatment of post-LBO asset sales is deliberately conservative: assets are assigned to their eventual owners. This prevents local divestitures from being misread as new entry.
- This matters because one prominent story about Safeway’s LBO was that it shed weak divisions. The paper is designed to avoid mechanically mistaking those divestitures for changes in rivalry.

4.4 Why early versus late LBOs help identification

- Expansion by incumbents should occur faster than de novo entry, because incumbents already know the market and have distribution infrastructure.
- The data line up with this timing prediction: incumbent expansion responds not only to early LBOs but also roughly similarly to later ones, while de novo entry responds mainly to earlier LBOs.
- This timing pattern is difficult to reconcile with a simple “chronically weak firms” interpretation.

4.5 Why the weak-firm alternative is not persuasive

- **Accounting evidence.** Future LBO firms are not significantly worse in operating income/sales, net income/sales, market-to-book assets, capital expenditures/assets, retained earnings/net income, or dividends/net income.
- **Event-study evidence.** If the LBO only revealed a firm was weak, rival returns would not necessarily jump on announcement; if leverage made the firm tougher, rival returns should fall. Instead, they rise.

- **Store timing evidence.** Non-LBO incumbents expand in the presence of both early and late LBO rivals, but entry is mainly tied to early LBOs, consistent with different response lags after a genuine shift in rivalry.

4.6 Relation to the literature

- The paper speaks to theories of capital structure and product-market competition.
- The evidence supports the “softer competition” models of Fudenberg and Tirole (1986), Bolton and Scharfstein (1990), and Phillips (1991).
- The evidence goes against the “tougher competition” predictions of Brander and Lewis (1986), Maksimovic (1988), and Rotemberg and Scharfstein (1990).
- Methodologically, the paper also draws on merger-style event studies that examine the stock-price response of rivals.

5 Tables

5.1 Tables 1 and 2: event setup and rival sample

TABLE 1—EVENTS INCLUDED IN THE EVENT ANALYSIS

Event	First announcement	Date	Date final transaction announced
Safeway LBO	Dart Group announces that it has acquired a 6-percent stake in Safeway	6/13/86	7/29/86
Supermarkets General LBO	Dart Group proposes to buy Supermarkets General	3/10/87	4/23/87
Stop & Shop LBO	Dart Group announces that it seeks a major stake in Stop & Shop	1/15/88	3/1/88
Kroger leveraged recap	The Haft family (who control Dart Group) reveals that it has a major stake in Kroger	9/13/88	10/10/88

Note: The event window is from 30 days prior to the first announcement, through the announcement of the final transaction.

TABLE 2—RIVAL SUPERMARKET CHAINS INCLUDED IN THE EVENT STUDY

Albertsons
American Stores
Brunos
Delchamps
Food Lion
Foodarama
Giant Food Stores
Great Atlantic and Pacific Tea Company
Hannaford Brothers
Marsh Supermarkets
Ruddick
Weis Markets
Winn-Dixie

Read:

- These tables define the event-study design.
- Table 1 lists the four major leverage events and their announcement windows.
- Table 2 lists the 13 rival supermarket chains in the event-study sample.
- The important point is that the event study is built around the largest and most visible supermarket leverage shocks of the period.

5.2 Table 3: rival stock-price responses to LBO announcements

Read:

- Direct competitors have **positive abnormal returns** for all four events.
- The Safeway coefficient is about 0.00317 and statistically significant at the 1% level; the Kroger coefficient is about 0.00186 and significant at the 5% level.
- Noncompeting firms show little systematic response, and the joint hypothesis that all four noncompeting-event coefficients are zero cannot be rejected even at loose significance levels.
- This is the paper's cleanest evidence that leverage is expected to **soften local product-market competition** rather than merely trigger general takeover speculation.

TABLE 3—EVENT COEFFICIENTS

Event	Event coefficient for other supermarket chains	
	(1) Competing	(2) Not competing
Safeway LBO	0.003168** (0.000966)	0.001614 (0.001044)
Supermarkets	0.001782 (0.001141)	−0.000175 (0.000817)
General LBO	0.001573 (0.001381)	−0.000082 (0.000793)
Stop & Shop LBO	0.001857* (0.000930)	0.000991 (0.001229)
Kroger leveraged recapitalization		

Notes: Column 1 reports event coefficients for firms competing in some of the same MSA's as the "event" firm; column 2 reports event coefficients for firms competing in none of the same MSA's as the "event" firm. The coefficients were estimated using seemingly unrelated regressions. Standard errors are given in parentheses.

*Statistically different from zero at the 5-percent level.

**Statistically different from zero at the 1-percent level.

TABLE 4—SUMMARY STATISTICS

Variable	Description	Mean	Standard deviation
A. Variables:			
Change in households (10,000's) [percentage change households]	Change in the number of households in the MSA between 1985 and 1991	5.59 [12.0]	5.93 [10.1]
Change in households per square mile [percentage change households per square mile]	Change in households per square mile in the MSA between 1985 and 1991. This is included because the change in households may have a different impact if spread over a very large or very small area.	49.7 [0.01]	157 [0.03]
Change in median income (\$10,000's) [percentage change in median income]	Change in median income in the MSA between 1985 and 1991	1.53 [44.2]	4.73 [16.5]
Change in median income squared (in \$1 × 10 ⁸) [percentage change in squared income]	Change in squared median income in the MSA between 1985 and 1991	9.92 [172]	4.66 [51.7]
Change in the share households with income less than \$10,000	Change in the share of households with annual incomes of less than \$10,000 in the MSA between 1985 and 1991	−0.0534	−0.0227
Deviation in mean stores per household [percentage deviation in mean stores per household]	MSA's deviation in 1985 from the number of stores that it would be predicted to have given the number of households in the MSA in 1985. I estimate that MSA's in 1985 have 40 stores plus 2.3 × 10 ^{−4} stores per household.	0.00 [−7.11]	30.0 [24.4]
Share of LBO firms	Share of stores in the MSA in 1985 owned by firms that would undertake LBO's by 1991	0.220	0.191
Share of early-LBO firms	Share of stores in the MSA in 1985 owned by firms that would undertake LBO's prior to 1988	0.108	0.156
Share of late-LBO firms	Share of stores in the MSA in 1985 owned by firms that would undertake LBO's in 1988 or later	0.111	0.128
Herfindahl index	Sum of the squared market shares of the five firms with the largest market shares in the MSA, where a firm's market share is defined as its share of total stores in a market	0.120	0.020
B. Incumbent Firm Variables:			
Store share, non-LBO incumbents	Incumbent firm's share of the total stores in the MSA in 1985	0.116	0.098
Store share, LBO incumbents	Incumbent firm's share of the total stores in the MSA in 1985	0.143	0.108
Total stores in chain, non-LBO incumbents	Total stores in the sample of 100 MSA's owned by the incumbent in 1985	307	284
Total stores in chain, LBO incumbents	Total stores in the sample of 100 MSA's owned by the incumbent in 1985	283	227

TABLE 4—Continued

B. Incumbent Firm Variables: (Continued)	
Number of firm-MSA observations in which non-LBO incumbents add stores:	79
Number of firm-MSA observations in which non-LBO incumbents neither add nor subtract stores:	20
Number of firm-MSA observations in which non-LBO incumbents subtract stores:	85
Number of firm-MSA observations in which LBO incumbents add stores:	47
Number of firm-MSA observations in which LBO incumbents neither add nor subtract stores:	19
Number of firm-MSA observations in which LBO incumbents subtract stores:	47
Number of MSA's in which no store entry occurs:	39

5.3 Table 4: variable definitions and summary statistics

Read:

- Table 4 lays out the main market-level and firm-level variables used in the entry and expansion regressions.
- It makes clear that the paper controls for demographic changes, initial over- or under-storing, and local concentration.
- It also reveals that the average LBO share across cities is economically meaningful rather than trivial.

5.4 Table 5: full-market OLS regressions

Read:

- When all LBO firms are pooled together, the coefficient on the 1985 LBO share is positive but not statistically significant at standard levels.
- Once LBOs are split by timing, the share of **early LBO firms** has a positive coefficient of about 0.174, significant at roughly the 7% level, while the share of late LBO firms is near zero.

- Economically, the author interprets the baseline magnitude as implying that if a firm with about 10% market share undertakes an LBO, total store growth in the market rises by roughly 1% more than otherwise.
- This is suggestive, but it is not the paper's strongest result because total market structure may adjust slowly.

TABLE 5—OLS SPECIFICATIONS

Variable	Coefficients		
	(1)	(2)	(3)
Constant	-0.0070 (0.1386)	0.0075 (0.1380)	-0.0006 (0.1572)
Percentage change in households	0.6347** (0.1569)	0.5952** (0.1581)	0.5377** (0.1651)
Percentage change in income	-0.6731 (1.1708)	-0.7661 (1.6374)	-0.5778 (1.1935)
Percentage change in income squared	0.1974 (0.3558)	0.2282 (0.3537)	0.1918 (0.3630)
Change in share with income less than \$10,000	-0.0071 (0.0084)	-0.0076 (0.0084)	-0.0044 (0.0088)
Percentage change in households per square mile	58.7209 (54.0745)	60.0863 (53.6788)	52.8402 (54.2677)
Percentage deviation from mean stores per household	-0.0866 (0.0575)	-0.0831 (0.0571)	-0.1194 (0.0620)
Herfindahl index	-0.1079 (0.2818)	-0.1171 (0.2798)	-0.2920 (0.3036)
Share LBO	0.0966 (0.0775)	—	—
Share early LBO	—	0.1736* (0.0931)	0.1438 (0.1369)
Share late LBO	—	-0.0175 (0.1094)	0.0280 (0.1141)
Regional dummies included?	no	no	yes
R^2	0.30	0.32	0.34
N	85	85	85

Notes: The dependent variable is the percentage change in the total number of stores in the MSA between 1985 and 1991. Standard errors are in parentheses.
 *Significantly different from zero at the 5-percent level.
 **Significantly different from zero at the 1-percent level.

TABLE 6—MAXIMUM-LIKELIHOOD ESTIMATION RESULTS FOR INCUMBENT FIRMS

Variables	A. Non-LBO incumbents			B. LBO incumbents		
	Coefficient	Marginal effects		Coefficient	Marginal effects	
		$d \Pr[y = -1]$ dx	$d \Pr[y = 1]$ dx		$d \Pr[y = -1]$ dx	$d \Pr[y = 1]$ dx
Change in households	0.0339* (0.0198)	-0.0134	0.0129	-0.0262 (0.0262)	0.0102	-0.0054
Change in income	-2.7210 (2.0230)	1.0788	-1.0358	3.9310 (2.6440)	-1.5337	0.8153
Change in income squared	0.2590 (0.1850)	-0.1027	0.0986	-0.3690 (0.2400)	0.1440	-0.0765
Change in share with income less than \$10,000	-0.2128 (0.1435)	0.0844	-0.0810	0.2154 (0.1764)	-0.0840	0.0447
Change in households per square mile	-0.0008 (0.0006)	0.0003	-0.0003	0.0005 (0.0063)	-0.0002	0.0001
Deviation from mean stores per household	0.0010 (0.0038)	-0.0004	0.0004	0.0032 (0.0046)	-0.0012	0.0007
Total stores	-0.0012** (0.0004)	0.0005	-0.0004	0.0007 (0.0006)	-0.0003	0.0001
Market share	2.1970* (1.0566)	-0.8711	0.8363	-2.7866* (1.5580)	1.0872	-0.5780
Herfindahl index	2.1396 (1.7995)	-0.8483	0.8145	0.1315 (3.6680)	-0.0513	0.0273
Share LBO	1.7016* (0.7557)	-0.6746	0.6477	1.7620 (1.1292)	-0.6875	0.3655
Exit threshold	0.0996 (0.6693)			0.8998 (0.9534)		
Entry threshold	0.4173 (0.6697)			1.3547 (0.9561)		
Number of observations	184			113		

Notes: The dependent variable has the following values: $Y_{ij} = +1$ if firm i withdraws at least one store from market j , $Y_{ij} = 0$ if firm i neither adds nor withdraws stores from market j , and $Y_{ij} = +1$ if firm i adds stores in market j . Standard errors are reported in parentheses.
 *Significantly different from zero at the 10-percent level.
 *Significantly different from zero at the 5-percent level.
 **Significantly different from zero at the 1-percent level.

5.5 Table 6: incumbent expansion decisions

Read:

- For **non-LBO incumbents**, the share of rival stores owned by LBO firms has a positive and statistically significant coefficient in the ordered probit.
- The paper's marginal-effect calculation implies that, holding other market characteristics at their mean, an LBO by a firm with a 10% market share raises the probability that a non-LBO incumbent adds stores by about **6.5%**.
- For **LBO incumbents**, the coefficient is also positive but estimated less precisely.
- The main message is that rivals already in the market react by expanding, which is exactly what one would expect if leverage softened competition.

5.6 Table 7: early versus late LBOs in incumbent expansion

Read:

- For non-LBO incumbents, both early and late LBO exposure have positive coefficients, and the hypothesis that the early-LBO effect exceeds the late-LBO effect is not rejected.
- This is sensible because incumbent expansion can respond relatively quickly.

- The timing pattern therefore looks different from de novo entry, which should take longer.

TABLE 7—MAXIMUM-LIKELIHOOD ESTIMATION RESULTS FOR INCUMBENT FIRMS

Variables	A. Non-LBO incumbents			B. LBO incumbents		
	Marginal effects			Marginal effects		
	Coefficient	$d \Pr[y = -1]$ dx	$d \Pr[y = 1]$ dx	Coefficient	$d \Pr[y = -1]$ dx	$d \Pr[y = 1]$ dx
Change in households	0.0330 ^a (0.0202)	-0.0131	0.0126	-0.0283 (0.0265)	0.0110	-0.0057
Change in income	-2.6630 (2.0190)	1.0553	-1.0152	4.2400 (2.6840)	-1.6511	0.8565
Change in income squared	0.2550 (0.1850)	-0.1010	0.0972	-0.3950 (0.2430)	0.1538	-0.0798
Change in share with income less than \$10,000	-0.2125 (0.1438)	0.0842	-0.0810	0.2358 (0.1793)	-0.0918	0.0476
Change in households per square mile	-0.0008 (0.0006)	0.0003	-0.0003	0.0008 (0.0064)	-0.0003	0.0002
Deviation from mean stores per household	0.0012 (0.0038)	-0.0005	0.0004	0.0032 (0.0046)	-0.0013	0.0007
Total stores	-0.0012 ^{**} (0.0004)	0.0005	-0.0004	0.0006 (0.0006)	-0.0002	0.0001
Market share	2.1043 ^a (1.0503)	-0.8339	0.8022	-2.7507 ^a (1.5434)	1.0712	-0.5557
Herfindahl index	2.0584 (1.8184)	-0.8157	0.7847	-0.2282 (5.6914)	0.0889	-0.0461
Share early LBO	1.7032 ^a (0.9073)	-0.6749	0.6493	1.9126 (1.1192)	-0.7448	0.3864
Share late LBO	1.4087 (0.8865)	-0.5582	0.5370	1.13798 (1.3221)	-0.5373	0.2787
Exit threshold	0.0996 (0.6693)			0.9035 (0.9542)		
Entry threshold	0.4173 (0.6697)			1.3866 (0.9569)		
Number of observations	184			113		

Notes: The dependent variable has the following values: $Y_{ij} = -1$ if firm i withdraws at least one store from market j , $Y_{ij} = 0$ if firm i neither adds nor withdraws stores from market j , and $Y_{ij} = +1$ if firm i adds stores in market j . Standard errors are reported in parentheses.
^aSignificantly different from zero at the 10-percent level.
^{*}Significantly different from zero at the 5-percent level.
^{**}Significantly different from zero at the 1-percent level.

TABLE 8—OLS ESTIMATION RESULTS FOR INCUMBENT FIRMS

Variables	Coefficients	
	(1) Non-LBO incumbents	(2) LBO incumbents
Change in households	-0.0660 (0.1170)	-0.2690 (0.2350)
Change in income	0.2860 (6.2420)	14.0750 (14.1650)
Change in income squared	0.0900 (0.5750)	-1.3100 (1.2500)
Change in share with income less than \$10,000	-0.4167 (0.4084)	0.5441 (0.9351)
Change in households per square mile	-0.0033 (0.0039)	0.0003 (0.0352)
Deviation from mean stores per household	0.0575 [*] (0.0250)	0.0268 (0.0435)
Total stores	-0.0055 [*] (0.0022)	-0.0005 (0.0042)
Market share	7.0556 (5.4637)	-28.1176 ^{**} (10.2838)
Herfindahl index	29.6984 (11.9950)	30.6566 (27.0131)
Share, early LBO	7.9665 ^a (4.1466)	0.7648 (5.7235)
Share, late LBO	-1.0105 (3.9945)	-5.8049 (9.0537)
Constant	-6.4550 [*] (3.1960)	-3.2238 (6.7199)
R^2	0.12	0.14
Number of observations	184	113

Notes: The dependent variable is the number of stores that firm i has added to market j . It is negative if the firm has subtracted stores from market j . White (1980) robust standard errors are in parentheses.

^aSignificantly different from zero at the 10-percent level.

^{*}Significantly different from zero at the 5-percent level.

^{**}Significantly different from zero at the 1-percent level.

5.7 Table 8: OLS robustness for incumbent store changes

Read:

- Using the *number* of stores added or subtracted as the dependent variable instead of the ordered category leaves the basic conclusion intact for non-LBO incumbents.
- The coefficient on the **early-LBO share** remains positive and significant for non-LBO incumbents.
- There is little support for the idea that LBO incumbents respond strongly to rivals' LBOs.
- This table reinforces that the core response is rival expansion by non-LBO firms.

5.8 Table 9: de novo entry into local markets

Read:

- Entry occurs in 39 of the 85 markets.
- The coefficient on the **early-LBO share** is positive and statistically significant: about 2.42 in the probit, with a marginal effect around 0.363.
- The coefficient on the late-LBO share is positive but very imprecisely estimated.
- This timing difference fits the paper's logic: new entry is slower than incumbent expansion, so only earlier leverage shocks have had enough time to attract outside chains.

TABLE 9—RESULTS FOR NEW ENTRY

Variable	(1)	(2)	(3)
	Coefficient	Marginal effects $d \Pr[y = 1]$ dx	Coefficient
Change in households	0.0841* (0.0395)	0.0126	2.23×10^{-6} * (1.13×10^{-6})
Change in income	4.6880 (2.9970)	0.7038	1.15×10^{-4} (7.0×10^{-5})
Change in income squared	-0.4950 ^a (0.2910)	-0.0743	1.22×10^{-9a} (6.75×10^{-10})
Change in share with income less than \$10,000	0.2859 (0.1806)	0.0429	0.0702 (0.0388)
Change in households per square mile	-0.0025 (0.0032)	-0.0004	4.19×10^{-4} (3.97×10^{-4})
Deviation mean stores per household	0.0012 (0.0064)	0.0002	6.95×10^{-4} (0.0021)
Herfindahl index	0.0887 (2.9218)	0.0133	0.0147 (1.0290)
Share, early LBO's	2.4183 ^a (1.1330)	0.3630	0.7904 ^a (0.3835)
Share, late LBO's	0.6756 (1.2177)		0.2960 (0.4317)
Constant	-1.518 (1.0364)		0.0758 (0.3248)
R^2			0.2010

Notes: The first column shows probit results. The dependent variable has the following values: $Y_j = 0$ if no entry occurs in market j ; $Y_j = 1$ if entry occurs. The second column shows the marginal effects implied by the coefficients in column 1. The third column shows the results of the linear probability specification. Entry occurs in 39 of the 85 markets. Standard errors are reported in parentheses.

^aSignificantly different from zero at the 10-percent level.

^aSignificantly different from zero at the 5-percent level.

TABLE 10—COMPARISON OF 1985 ACCOUNTING VALUES FOR FIRMS THAT WOULD EVENTUALLY UNDERTAKE AN LBO AND FIRMS THAT WOULD NOT EVENTUALLY UNDERTAKE AN LBO

Accounting ratio	Mean		t statistic of difference
	LBO firm	Non-LBO firm	
Operating income/sales	0.0363	0.0395	0.48
Net income/sales	0.0040	0.0043	0.14
Market value/book value of assets	0.8316	0.8194	0.10
Capital expenditures/assets	0.1461	0.1300	0.80
Retained earnings/net income	0.3703	0.3266	0.76
Dividends/net income	0.2375	0.1896	0.39

5.9 Table 10: are future LBO firms simply weak firms?

Read:

- Future LBO and non-LBO firms look very similar in 1985 accounting data.
- There is no significant difference in operating income/sales, net income/sales, market-to-book assets, capital expenditures/assets, retained earnings/net income, or dividends/net income.
- So the main results are **not well explained** by the idea that LBO firms were simply obvious underperformers before their deals.

6 Short summary

- **Conclusion.** In the supermarket industry, large leverage increases are associated with softer local product-market competition.
- **How the paper shows it.** Rivals' stock prices rise when a competitor announces an LBO, and markets with more LBO incumbents attract more rival expansion and, with a lag, more entry.
- **Why this is credible.** The paper compares direct competitors with noncompetitors, adjusts carefully for post-LBO asset sales, and rejects the idea that future LBO firms were simply weak firms all along.
- **Big takeaway.** Capital structure matters not only for firm finance but also for the structure and aggressiveness of competition in product markets.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes an empirical link between leverage and product-market competition.
- In this setting, increased leverage appears to make competition **less tough**, not more.
- The result is visible both in financial markets (through rivals' announcement returns) and in real market outcomes (through store expansion and entry).

7.2 Economic intuition

- When a rival becomes highly levered, market participants appear to expect that it will compete less aggressively.
- If the rival is expected to compete less aggressively, incumbent competitors benefit immediately, which shows up in stock prices.
- Over time, those incumbents exploit the more favorable competitive environment by expanding, and outside chains eventually find it worthwhile to enter.

7.3 Takeaway

This paper is best understood as an early reduced-form test of whether financing decisions spill over into product markets. Its answer is yes. In the supermarket industry of the late 1980s, leveraged buyouts did not merely rearrange liabilities; they changed how firms behaved in local competition. The evidence points toward softer rivalry after leverage increases, consistent with theories in which debt weakens competitive aggressiveness. Even though the paper cannot pin down whether the change operates through prices, quality, or both, it makes a durable point: capital structure can reshape market structure.